# t2-w09 — condition C2

Simplification of Algebraic Terms — Simplification of algebraic expressions: order of operations, brackets and word problems

> C1 plus the lesson note as source evidence.

## INPUT LAYERS

| Input layer | What it carries | Channel | In this condition |
|---|---|---|---|
| Scope | what is to be taught, and to whom | prompt | **present** |
| Prior knowledge | what the learner already knew coming in | prompt | **present** |
| Materials | what the lesson was represented with | prompt | **present** |
| Position | where this objective sat in the sequence | prompt | **present** |
| Method | the shape of the teaching, as event labels | prompt | **present** |
| Enacted content | what was actually explained and worked through | prompt | **present** |
| Set work | what the learners were asked to do in class | prompt | **present** |
| Assessment | the questions the lesson was checked with | prompt | **present** |
| Figures | the drawings the lesson used | tool | absent |

## WHAT THE MODEL IS SENT

Contract `ComposeWriteGroundedSection`, declaring: `artefact_spec, lesson_context, unit_context, lesson_content`.

### Lesson context — shared by every section

**`subject`**

Mathematics

**`grade_level`**

JSS1

**`topic`**

Simplification of Algebraic Terms

**`subtopic`**

Simplification of algebraic expressions: order of operations, brackets and word problems

**`term`**

Second Term

**`week`**

9

**`lesson_objectives`**

- carry out order of operation in algebra
- insert/remove bracket and simplify expression
- solve word problems involving algebraic expression

**entry_behaviour**

- Addition and subtraction of algebraic terms
- Multiplication and division of algebraic terms

**instructional_materials**

- Chart showing algebraic expression.

### Objective 1 — carry out order of operation in algebra

**objective_text**

carry out order of operation in algebra

**current_step**

1

**instructional_events**

- provide_learning_guidance
- elicit_performance
- provide_feedback

**student_tasks**

- 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$

**assessment_points**

- Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$
- Simplify $7 \times 3a - (3a + 5a) \times 2$

**lesson_content**


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Example 1. Find the value of the following\n\n(a) $18 − 6 \\times 2$\n\n(b) $6
− 18 \\div 3$\n\n(c) $6 \\times 2 − 2 − 40 \\div 4$\n\n**Solution**\n\n(a) $18 − 6 \\times 2$
\n\n$= 18 − 12$\n\n$= 6$\n\n(b) $6 − 18 \\div 3$\n\n$= 6 − 6$\n\n$= 0$\n\n(c) $(6 \\times 2) − 2
− (40 \\div 4)$\n\n$= 12 − 2 − 10$\n\n$= 10 − 10$\n\n$= 0$\n\n#### Example 2. Simplify the
following\n\n(a) $3x \\times 2 + 5x$\n\n(b) $17y − 5 \\times 2y$\n\n**Solution**\n\n(a) $3x \
\times 2 + 5x$\n\n$= 6x + 5x$\n\n$= 11x$\n\n(b) $17y − 5 \\times 2y$\n\n$= 17y − 10y$\n\n$= 7y$
\n\n#### Example 3. Simplify the following\n\n(a) $7x \\div x + 5$\n\n(b) $4x − (6 \\div 3)$
\n\n(c) $(24x \\div 6) + x + (1 \\times 5x)$\n\n**Solution**\n\n(a) $7x \\div x + 5$\n\n$= \
\frac{7x}{x} + 5$\n\n$= 7 + 5$\n\n$= 12$\n\n(b) $4x − (6 \\div 3)$\n\n$= 4x − 2$\n\n(c) $(24x \
\div 6) + x + (1 \\times 5x)$\n\n$= 4x + x + 5x$\n\n$= 5x + 5x$\n\n$= 10x$\n\n#### Activity.
Simplify the following\n\n1. $4n + 3n \\times 10$\n2. $32 + 8x \\div 2$\n3. $3 \\times 5x + 4x \
\div 2$\n4. $36y \\div 6 + y + 2 \\times 4y$"
 }
]
```


**Withheld from this payload:** `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

### Objective 2 — insert/remove bracket and simplify expression

**objective_text**

insert/remove bracket and simplify expression

**current_step**

2

**instructional_events**

• provide_learning_guidance
• elicit_performance
• provide_feedback

**student_tasks**

• (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$

**assessment_points**

• Simplify $12 - (7 - 2f) - (3 - f)$
• Simplify $(8a - 5) + (14 - 3a)$

**lesson_content**


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Hints**\n\n- Always try to simplify the terms inside bracket first.\n- A $+$ sign
before a bracket does not change the signs of the terms inside the bracket, when\n  bracket is
removed.\n- A $-$ sign before a bracket will change all the $+$ sign inside the bracket to $-$
and also\n  change all the $-$ sign inside the bracket to $+$ when the bracket is removed.
\n\n#### Example 1. Simplify\n\n(a) $a + (b + c)$\n\n(b) $a + (b - c)$\n\n(c) $2a - (3b - c)$
\n\n**Solution**\n\n(a) $a + (b + c) = a + b + c$\n\n(b) $a + (b - c) = a + b - c$\n\n(c) $2a -
(3b - c) = 2a - 3b + c$\n\n#### Example 2. Simplify\n\n(a) $a + (a - b)$\n\n(b) $6x + (3 - x)$
\n\n(c) $(3a + 2) + (5a + a)$\n\n**Solution**\n\n(a) $a + (a - b)$\n\n$= a + a - b$\n\n$= 2a - b$
\n\n(b) $6x + (3 - x)$\n\n$= 6x + 3 - x$\n\n$= 6x - x + 3$\n\n$= 5x + 3$\n\n(c) $(3a + 2) + (5a +
a)$\n\n$= 3a + 2 + 5a + a$\n\n$= 3a + 5a + a + 2$\n\n$= 9a + 2$\n\n#### Example 3. Simplify the
following\n\n(a) $5x - (3y + z)$\n\n(b) $10d - (9c - 4d)$\n\n(c) $(6x - y) - (4x - 5y)$
\n\n**Solution**\n\n(a) $5x - (3y + z)$\n\n$= 5x - 3y - z$\n\n(b) $10d - (9c - 4d)$\n\n$= 10d -
9c + 4d$\n\n$= 10d + 4d - 9c$\n\n$= 14d - 9c$\n\n(c) $(6x - y) - (4x - 5y)$\n\n$= 6x - y - 4x +
5y$\n\n$= 6x - 4x + 5y - y$\n\n$= 2x + 4y$\n\n$= 2(x + 2y)$\n\n#### Students activity\n\n(a)
$(12a - 7) + (4 - 5a)$\n\n(b) $(8x - 2y) - (9x - 4y)$"
 }
]
```


**Withheld from this payload:** `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

### Objective 3 — solve word problems involving algebraic expression

**objective_text**

solve word problems involving algebraic expression

**current_step**

3

**instructional_events**

• provide_learning_guidance
• elicit_performance
• provide_feedback

**student_tasks**

• 1. A pen cost $h$ naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost $d$ naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.

**assessment_points**

• A pen and a pencil altogether cost $x$ naira. If a pen cost $y$ naira, find the cost of 3 pencils.

**lesson_content**


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Example 1\n\nA pen cost $P$ naira and a pencil cost 15 naira less than a pen.
\n\n(a) Find the cost of a pencil\n\n(b) Find the cost of a pen and a
pencil\n\n**Solution**\n\nCost of a pen $= ₦P$\n\n(a) cost of a pencil $= ₦(P − 15)$\n\n(b) So
cost of a pen and a pencil $= ₦P + ₦(P − 15)$\n\n$= ₦P + ₦P − ₦15$\n\n$= ₦2P − ₦15$\n\n$= ₦(2P −
15)$\n\n#### Example 2\n\nA pencil cost $x$ naira and a rubber cost 40 naira more than a pencil.
\n\n(a) find the cost of a rubber\n\n(b) find the cost of a pencil and a
rubber\n\n**Solution**\n\n(a) cost of a pencil $= ₦x$\n\n$\\therefore$ cost of a rubber $= ₦(x +
40)$\n\n(b) cost of a pencil and a rubber:\n\n$= ₦x + ₦(x + 40)$\n\n$= ₦(x + x + 40)$\n\n$= ₦(2x
+ 40)$\n\n#### Example 3\n\nIn a class, there are $x$ boys and $y$ girls last semester. This
semester, 5 boys join and\n3 girls leave the class. How many students are now in this class?
\n\n**Solution**\n\nTotal no $=$ number of boys $+$ number of girls\n\nNumber of boys $= x + 5$
\n\nNumber of girls $= y − 3$\n\nTotal number $= x + 5 + y − 3$\n\n$= x + y + 5 − 3$\n\n$= x + y
+ 2$\n\n#### Student activity\n\n1. A pen cost $h$ naira and a crayon cost ₦20 less than a pen.
\n   (a) Find the cost of a crayon\n   (b) Find the cost of a crayon and a pen\n2. Envelopes cost
$d$ naira per pack, 4 note pad cost 20 naira more than a pack of envelopes.\n   Find the total
cost of a pack of envelope and notepad."
 }
]
```


**Withheld from this payload:** `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

## WHAT THE MODEL PRODUCED

### Simplifying Algebraic Expressions – Order of Operations

> EXPLANATION
> When you simplify an algebraic expression you must follow the same order of operations you use with numbers:
>
> 1. **Brackets** – do anything inside brackets first.
> 2. **Multiplication and Division** – work from left to right.
> 3. **Addition and Subtraction** – work from left to right.
>
> Remember that multiplication and division are of equal priority, as are addition and subtraction. Always move from left to right within each level.
>
> For example, in the expression $18 - 6 \times 2$ you first multiply $6 \times 2 = 12$ and then subtract: $18 - 12 = 6$.
>
> If brackets are present, simplify inside them before anything else, e.g. $4x - (6 \div 3)$ becomes $4x - 2$ because the division inside the brackets is done first.

> WORKED_EXAMPLE
> ### Worked Example 1
> **Simplify** $18 - 6 \times 2$
>
> - Multiply first: $6 \times 2 = 12$
> - Subtract: $18 - 12 = 6$
>
> **Answer:** $6$
>
> ### Worked Example 2
> **Simplify** $6 - 18 \div 3$
>
> - Divide first: $18 \div 3 = 6$
> - Subtract: $6 - 6 = 0$
>
> **Answer:** $0$
>
> ### Worked Example 3
> **Simplify** $6 \times 2 - 2 - 40 \div 4$
>
> - Multiply and divide left-to-right:
>  - $6 \times 2 = 12$
>  - $40 \div 4 = 10$
> - Now the expression is $12 - 2 - 10$
> - Subtract left-to-right: $12 - 2 = 10$, then $10 - 10 = 0$
>
> **Answer:** $0$

**1.** Simplify the expression $4n + 3n \times 10$.
answer: $34n$

**2.** Simplify the expression $3 \times 5x + 4x \div 2$.
answer: 15x + 2x = 17x$

**3.** Simplify the expression $36y \div 6 + y + 2 \times 4y$.
answer: $6y + y + 8y = 15y$

**4.** Simplify $14pq \times r$.
answer: $14pqr$

**5.** Simplify $12x^2y \div 3x$.
answer: $4xy$

**6.** Simplify $7 \times 3a - (3a + 5a) \times 2$.
answer: $-8a$

**7.** Simplify the expression $32 + 8x \div 2$.
answer: $32 + 4x$

## Removing Brackets and Simplifying Algebraic Expressions

EXPLANATION
### How to Remove Brackets

- **First, simplify the terms inside the brackets.**
- A **+** sign before a bracket does **not** change the signs of the terms inside when the bracket is removed.
- A **–** sign before a bracket **reverses** the signs of every term inside the bracket.

**Remember:** work from the inside out and keep track of the signs.

WORKED_EXAMPLE
#### Example 1

(a) $a + (b + c)$

**Solution**
$a + (b + c) = a + b + c$

(b) $a + (b - c)$

**Solution**
$a + (b - c) = a + b - c$

(c) $2a - (3b - c)$

**Solution**
$2a - (3b - c) = 2a - 3b + c$

WORKED_EXAMPLE
#### Example 2

(a) $a + (a - b)$

**Solution**
$a + (a - b) = a + a - b = 2a - b$

(b) $6x + (3 - x)$

**Solution**
$6x + (3 - x) = 6x + 3 - x = 5x + 3$

(c) $(3a + 2) + (5a + a)$

**Solution**
$(3a + 2) + (5a + a) = 3a + 2 + 5a + a = 9a + 2$

WORKED_EXAMPLE
#### Example 3

(a) $5x - (3y + z)$

**Solution**
$5x - (3y + z) = 5x - 3y - z$

(b) $10d - (9c - 4d)$

**Solution**
$10d - (9c - 4d) = 10d - 9c + 4d = 14d - 9c$

(c) $(6x - y) - (4x - 5y)$

**Solution**
$(6x - y) - (4x - 5y) = 6x - y - 4x + 5y = 2x + 4y = 2(x + 2y)$

**1.** (a) $(12a - 7) + (4 - 5a)$
answer: 7a - 3

**2.** (b) $(8x - 2y) - (9x - 4y)$
answer: -x + 2y

**3.** Simplify $12 - (7 - 2f) - (3 - f)$
answer: 2 + 3f

**4.** Simplify $(8a - 5) + (14 - 3a)$
answer: 5a + 9

Objective 3

### Solving Word Problems with Algebraic Expressions

EXPLANATION
When a word problem gives the cost of one item and tells you how another item's cost relates to it, you can turn the words into an algebraic expression.

1. Identify the unknown quantity and give it a letter (e.g., let the cost of a pen be **P**).
2. Translate the relationship into an expression (e.g., a crayon costs **20** naira less than a pen → **P - 20**).
3. Use the expression to answer the question, adding or subtracting as required.

Remember to keep the units (naira) with each term and to combine like terms only after the expressions are written.

WORKED_EXAMPLE
**Example 1**
A pen costs **P** naira and a pencil costs 15 naira less than the pen.

(a) Cost of the pencil = **P - 15** naira.

(b) Cost of a pen and a pencil = **P + (P - 15) = 2P - 15** naira.

WORKED_EXAMPLE
**Example 2**
A pencil costs **x** naira and a rubber costs 40 naira more than the pencil.

(a) Cost of the rubber = **x + 40** naira.

(b) Cost of a pencil and a rubber = **x + (x + 40) = 2x + 40** naira.

WORKED_EXAMPLE
**Example 3**
There are **x** boys and **y** girls. This semester 5 boys join and 3 girls leave.

New number of boys = **x + 5**.
New number of girls = **y - 3**.
Total students = **(x + 5) + (y - 3) = x + y + 2**.

**1.** 1. A pen costs **h** naira and a crayon costs 20 naira less than the pen. (a) Find the cost of the crayon. (b) Find the total cost of the crayon and the pen.
answer: (a) **h - 20** naira (b) **h + (h - 20) = 2h - 20** naira

**2.** 2. Envelopes cost **d** naira per pack. A notepad costs 20 naira more than a pack of envelopes. Find the total cost of one pack of envelopes and one notepad.
answer: **d + (d + 20) = 2d + 20** naira

**3.** A pen and a pencil together cost **x** naira. If a pen costs **y** naira, find the cost of three pencils.
answer: Cost of one pencil = **x - y** naira, so three pencils cost **3(x - y)** naira.

## LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

| Output layer | Made | Written from | Verdict |
|---|---|---|---|
| Title<br>the reader-facing framing of the section | 3 | Scope | **grounded** |
| Explanation<br>the principle, rule or relationship to review | 3 | Enacted content | **grounded** |
| Worked example<br>a problem with its solution worked through | 7 | Enacted content | **grounded** |
| Figure<br>a drawing the learner reads from | 0 | nothing in the input | not produced |
| Practice<br>what the learner does | 14 | Set work | **grounded** |
| Answer<br>what the learner checks against | 14 | Enacted content | **grounded** |